

QUESTION BANK

Unit – II [Vector Space and Eigen Values]

Basic concepts involving Vector space, Dimension and Basis, Orthogonal bases and Gram-Schmidt Orthogonalization Eigen values and Eigen vectors of matrices, Similarity of matrices and Diagonalization.

LONG QUESTIONS

1.	Examine whether the following vectors are linearly independent. (i) $[1, -1, 1, 1], [1, 1, -1, 0], [-1, 1, 1, 1], [0, 1, 0, 0]$ (ii) $[7, -3, 11, -6], [-56, 24, -88, 48], [1, -3, 11, -6], [-56, 24, -88, 48]$ (iii) $[1/4, 0, -1/4], [0, 1/2, -1/2], [1/3, -1/3, 0]$ (iv) $[1/2, 2/3, 3/4, 4], [0, 0, 2, 0], [1, 2, 0, 1], [1/2, 2/3, 3/4, 4]$ (v) $\{x^2 - 1, x + 1, x - 1\}$ (vi) $\{x, x^3 - x, x^4 + x^2, x + x^2 + x^4 + 1/2\}$ (vii) $\{1 + 2x, x^2, 3 + 5x + 2x^2\}$ (viii) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
2.	Prove that the following sets are vector spaces and find their dimension and one basis. (i) Set of all $n \times m$ real matrices. (ii) Set of all complex numbers. (iii) Set of all n-tuples of real numbers (iv) Set of all $n \times n$ matrices (v) Set of all $n \times n$ symmetric matrices
3.	Using Gram-Schmidt process find an orthogonal basis and an orthonormal basis of the following vectors. (i) $(1, 1, 1, 1), (1, 1, -1, -1), (1, 2, 0, 2),$ (ii) $(1, 0, 1), (1, 1, 1), (1, 3, 4)$ (iii) $(1, 2, -2), (2, 0, 1), (1, 1, 0),$ (iv) $(1, 1, 0), (1, 0, 1), (0, 1, 1)$ (v) $(1, 0, 1, 1), (-1, 0, -1, 1), (0, -1, 1, 1)$
4.	Using Gram-Schmidt process find a QR factorization of the following matrices. (i) $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix}$ (iv) $\begin{bmatrix} 0 & 3 & 5 \\ 3 & 3 & 4 \\ 4 & 1 & 6 \end{bmatrix}$ (v) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$
	Find a vector x orthogonal to the row space of A , and a vector y orthogonal to the column space of A ,

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

5.	and a vector z orthogonal to the null space of A , where
6.	Let P be the plane in R^3 with equation $x + 2y - z = 0$. Find a vector perpendicular to P . What matrix has the plane P as its null space, and what matrix has P as its row space.
7.	Find all vectors that are perpendicular to $(1, 4, 4, 1)$ and $(2, 9, 8, 2)$.
8.	Let S be the subspace of R^4 containing all vectors with $x_1 + x_2 + x_3 + x_4 = 0$. Find a basis for the space $S^\perp$, containing all vectors orthogonal to S .
9.	Find the orthogonal complement of the plane spanned by the vectors $(1, 1, 2)$ and $(1, 2, 3)$ by taking these to be the rows of A and solving $Ax = 0$.
10.	Find a matrix whose row space contains $(1, 2, 1)$ and whose null space contains $(1, -2, 1)$ or prove that there is no such matrix.
11.	Find the eigen values and the set of all linearly independent eigenvectors of the following matrices. (i) $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$ (iii) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & 2 \\ 2 & -4 & 3 \end{bmatrix}$ (iv) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ (v) $\begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ a] Determine whether the eigen vectors are orthogonal. b] Determine the algebraic and geometric multiplicities of the eigen values.
12.	Prove the following. (i) The trace of a matrix is equal to the sum of the eigen values of the matrix. (ii) The determinant of the matrix is equal to the product of the eigen values of the matrix.
13.	Find a basis of eigen vectors and diagonalize the following matrices. Also find the square of the matrix. (i) $\begin{bmatrix} 7.3 & 0.2 & -3.7 \\ -11.5 & 1.0 & 5.5 \\ 17.7 & 1.8 & -9.3 \end{bmatrix}$ (ii) $\begin{bmatrix} 18 & 0 & 0 \\ 24 & -4 & 0 \\ 42 & -12 & 2 \end{bmatrix}$ (iii) $\begin{bmatrix} -2.5 & -3 & 3 \\ -4.5 & -4 & 6 \\ -6 & -6 & 8 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 2 & -9 \\ -2 & -4 & 19 \\ 0 & -1 & 2 \end{bmatrix}$ (v) $\begin{bmatrix} 5 & 10 & -10 \\ 10 & 5 & -20 \\ 5 & -5 & -10 \end{bmatrix}$
14.	If $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of an $n \times n$ matrix, then show that the corresponding eigenvectors $x_1, x_2, \dots, x_k$ form a linearly independent set.
15.	Is the given set of vectors a vector space? Give reasons. If your answer is yes, determine the dimension and find a basis? All vectors (v_1, v_2, v_3) in R^3 satisfying the condition $3v_1 - 2v_2 + v_3 = 0$.
16.	Is the given set of vectors a vector space? Give reasons. If your answer is yes, determine the dimension and find a basis? All vectors (v_1, v_2, v_3) in R^3 satisfying the condition $4v_2 + v_3 = 0$.
17.	Examine whether all vectors in R^n with first $n - 2$ components zero form a vector space. Determine the dimension and find a basis.

18.	If an $n \times n$ matrix A has a basis of eigen vectors, then show that $D = \vec{X}^{-1}A\vec{X}$ is a diagonal matrix with eigen values of A as the entries on the principal diagonal of D . Here $\vec{X}$ is a matrix with the eigen vectors of A as the column vectors. Also prove that for any positive integer $m \geq 2$, $D^m = \vec{X}^{-1}A^m\vec{X}$.
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SHORT QUESTIONS

1.	Examine whether the vectors $x_1 = [1, 9, 9, 8]$, $x_2 = [2, 0, 0, 3]$, $x_3 = [2, 0, 0, 8]$ are linearly independent.
2.	Examine whether the vectors $x_1 = [1, 9, 9, 8]$, $x_2 = [2, 0, 0, 3]$, $x_3 = [2, 0, 0, 8]$ are linearly dependent or independent.
3.	Examine whether the set of following vectors form a vector space. $(v_1, v_2, v_3, v_4, v_5)$ in R^5 with $v_1 = 0, v_3 = 0, v_2 + v_4 \geq 0$
4.	Find t so that $(1-t, 0, 0), (1, 1-t, 0)$ and $(1, 1, 1-t)$ are linearly dependent.
5.	If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find the eigenvalues of A^{-1} .
6.	If $\lambda = 0$ is an eigenvalue of the matrix A , then what one can say about $ A $.
7.	If A is any 5×5 matrix, and a polynomial associated with the matrix A is defined by $ \lambda I - A = \sum_{i=0}^n \alpha_i \lambda^i$, then what is the determinant of the matrix A .
8.	What is a basis of eigen vectors? When does it exist?
9.	Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 3 \\ 3 & 6 \end{bmatrix}$, find an eigenvector of $P^{-1}AP$.
10.	Show that eigen values of a unitary matrix have absolute value 1.
11.	Find the spectrum of eigen vectors of the matrix $\begin{bmatrix} 4 & 0 \\ 0 & -6 \end{bmatrix}$.
12.	Find the spectrum of eigen vectors of the matrix $\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$.
13.	Prove that the eigen values of a symmetric matrix are real numbers
14.	Prove that the eigen values of a skew -symmetric matrix are pure imaginary or zero.
15.	Show that eigen values of a orthogonal matrix have absolute value 1.
16.	Examine whether the matrix $\begin{bmatrix} 0.96 & -0.28 \\ 0.28 & 0.96 \end{bmatrix}$
17.	Show that the eigen values of a Hermitian matrix are real.
18.	Prove that the eigen values of a skew -Hermitian matrix are pure imaginary or zero.
19.	Show that eigen values of a unitary matrix have absolute value 1.
20.	Find the eigen values of the matrix A^3 if $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$.

21.	Examine whether the vectors $[1, -1, 1]$, $[1, 1, -1]$, $[-1, 1, 1]$, $[0, 1, 0]$ are linearly independent.
22.	For any square matrix A prove that A and A^T have the same eigen values.
23.	Let λ be an eigen value of a square matrix A then prove that λ^2 is an eigen value of the matrix A^2 .
24.	Let λ be an eigen value of a square matrix A then for any positive integer r , prove that λ^r is an eigen value of the matrix A^r .